

Lab — Build a News Digest Tool

Step 1 — Get sample articles

```
articles = [
    """Apple unveiled its newest iPhone at an event in Cupertino on
    Tuesday.
    CEO Tim Cook said the device features a faster chip and improved
    camera.
    The phone will be available in stores starting next month, the company
    said, with prices starting at $999.""",
    """The World Health Organization announced new guidelines on Monday
    aimed at reducing sugar consumption worldwide. Officials in Geneva
    said
    the changes could prevent millions of cases of diabetes over the next
    decade if adopted by national governments.""",
    """Tesla reported record deliveries for the third quarter, beating
    analyst expectations. The Texas-based company said demand remained
    strong in China and Europe despite increased competition from
    traditional automakers entering the EV market.""",
    """NASA's Perseverance rover discovered new evidence of ancient
    water activity on Mars, according to a paper published this week.
    Scientists at the Jet Propulsion Laboratory said the findings
    strengthen the case that the planet once had a wetter climate.""",
    """Manchester United announced the signing of a new midfielder
    from Barcelona for a reported fee of 80 million euros. The 24-year-old
    player will join the club ahead of the new season starting in
    August.""",
]
```

Step 2 — Load models once

```
import spacy
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Load each model ONCE — reused for every article
nlp = spacy.load('en_core_web_sm')

# Direct loading for summarization model
summarizer_tokenizer = AutoTokenizer.from_pretrained("sshleifer/distilbart-cnn-12-6")
summarizer_model = AutoModelForSeq2SeqLM.from_pretrained("sshleifer/distilbart-cnn-12-6")
```

Loading weights: 100%

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Step 3 — Process each article

```
def digest_article(text):
    # Summarise (abstractive, short)
    inputs = summarizer_tokenizer(text, return_tensors="pt", truncation=True, max_length=1024)
    summary_ids = summarizer_model.generate(inputs["input_ids"], num_beams=4, max_length=40, min_length=10, do_sample=False, ea
    summary_text = summarizer_tokenizer.decode(summary_ids[0], skip_special_tokens=True)

    # Extract entities
    doc = nlp(text)
    entities = {}

    for ent in doc.ents:
        entities.setdefault(ent.label_, set()).add(ent.text)

    return summary_text, entities

for i, article in enumerate(articles, 1):
    summary, entities = digest_article(article)

    print(f"--- Article {i} ---")
    print("Summary :", summary)
    print("Entities:")

    for label, names in entities.items():
```

```
print(f" {label}: {' '.join(names)}")

print()

--- Article 1 ---
Summary : CEO Tim Cook said the device features a faster chip and improved camera . The phone will be available in stores start
Entities:
  ORG: Apple
  GPE: Cupertino
  DATE: Tuesday, next month
  PERSON: Tim Cook
  MONEY: 999

--- Article 2 ---
Summary : The World Health Organization announced new guidelines aimed at reducing sugar consumption worldwide . Officials in G
Entities:
  ORG: The World Health Organization
  DATE: Monday, the next
decade
  GPE: Geneva
  CARDINAL: millions

--- Article 3 ---
Summary : Tesla reported record deliveries for the third quarter, beating analyst expectations . The Texas-based company said d
Entities:
  DATE: the third quarter
  GPE: China, Texas
  LOC: Europe
  ORG: EV

--- Article 4 ---
Summary : NASA's Perseverance rover discovered new evidence of ancient water activity on Mars . Scientists said the findings st
Entities:
  ORG: the Jet Propulsion Laboratory, NASA
  LOC: Mars
  DATE: this week

--- Article 5 ---
Summary : Manchester United announce signing of Barcelona midfielder from Barcelona . The 24-year-old midfielder will join the
Entities:
  PERSON: Manchester United
  GPE: Barcelona
  CARDINAL: 80 million
  DATE: 24-year-old, August
```

Start coding or [generate](#) with AI.